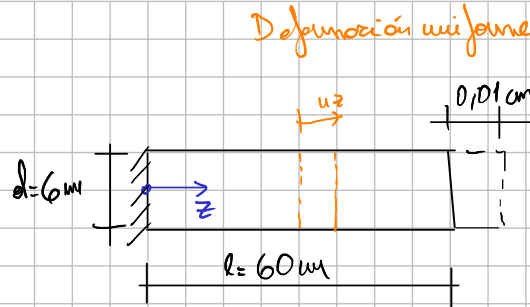
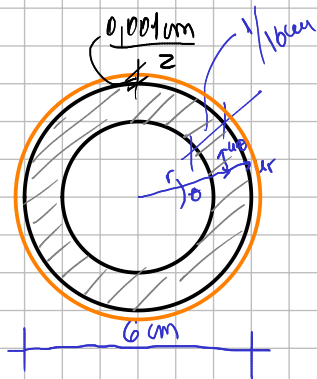
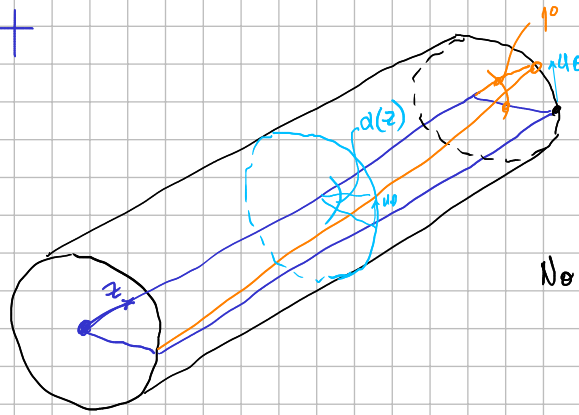




Deformación uniforme

Pequeñas deformaciones



No se produce abarrotamiento

- 1) Estirado axialmente 0,010 cm
- 2) Expansión en diámetro 0,001 cm
- 3) Giro de torsión 10

→ Desde los desplazamientos se quiere obtener las deformaciones.

$$e_{rr} = \frac{\partial u_r}{\partial r}, \quad \times$$

$$e_{\theta\theta} = \frac{u_r}{r} + \frac{1}{r} \frac{\partial u_\theta}{\partial \theta}, \quad \times$$

$$e_{r\theta} = \frac{1}{2} \left(\frac{1}{r} \frac{\partial u_r}{\partial \theta} + \frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r} \right), \quad \times$$

$$e_{zr} = \frac{1}{2} \left(\frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r} \right),$$

$$e_{z\theta} = \frac{1}{2} \left(\frac{1}{r} \frac{\partial u_z}{\partial \theta} + \frac{\partial u_\theta}{\partial z} \right), \quad \times$$

$$e_{zz} = \frac{\partial u_z}{\partial z}, \quad \times$$

• ϵ_{zz}

$$\epsilon_{zz} = \frac{\partial u_z}{\partial z} \Rightarrow u_z - u_z(0) = \int_0^l \epsilon_{zz} dz = \epsilon_{zz} z \Big|_0^l = \epsilon_{zz} l$$

$$\epsilon_{zz} = \frac{\Delta l}{l} = \frac{u_z - u_z(0)}{l} = \frac{0,01 \text{ cm} - 0 \text{ cm}}{60 \text{ cm}} = 1,67 \times 10^{-4}$$

• $\epsilon_{\theta\theta}$

$$\epsilon_{\theta\theta} = \frac{u_r}{r} + \frac{1}{r} \cdot \frac{\partial u_\theta}{\partial \theta}$$

→ no cambia con θ al girar como un círculo
 $\frac{\partial u_\theta}{\partial \theta} = 0$

$$\epsilon_{\theta\theta} = \frac{u_r}{r}$$

$$\epsilon_{\theta\theta e} = \frac{u_{re}}{r_e} = \frac{\frac{0,001 \mu m}{2}}{\frac{6 \mu m}{2}} = 1,67 \times 10^{-4}$$

$$\frac{\cancel{\pi} De - \cancel{\pi} de}{\cancel{\pi} de} = \frac{De - de}{de} = \frac{\cancel{2} R_e - \cancel{2} r_e}{2 r_e} = \frac{R_e - r_e}{r_e} = \frac{u_{re}}{r_e} = \epsilon_{\theta\theta}$$

$$\epsilon_{\theta\theta i} = \frac{u_{ri}}{r_i} = \frac{\frac{0,001 \mu m}{2}}{(6 \mu m - \frac{1 \mu m}{16})} = 1,70 \times 10^{-4}$$

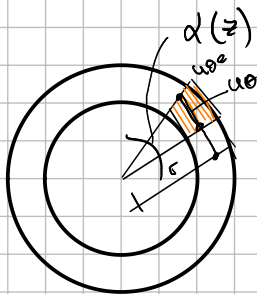
• ϵ_{rr}

$$\epsilon_{rr} = \frac{\Delta u_r}{\Delta r} = \frac{u_{re} - u_{ri}}{r_e - r_i} = \frac{0}{1/16 \mu m} = 0$$

longitud inicial

• $\epsilon_{r\theta}$

$$\epsilon_{r\theta} = \frac{1}{2} \left(\frac{1}{r} \frac{\partial u_r}{\partial \theta} + \frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r} \right)$$

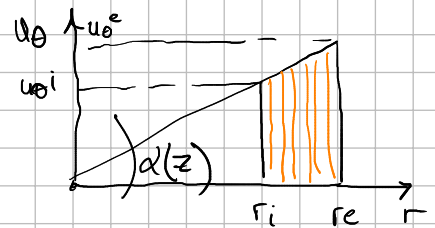
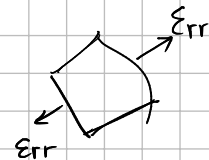
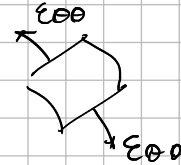
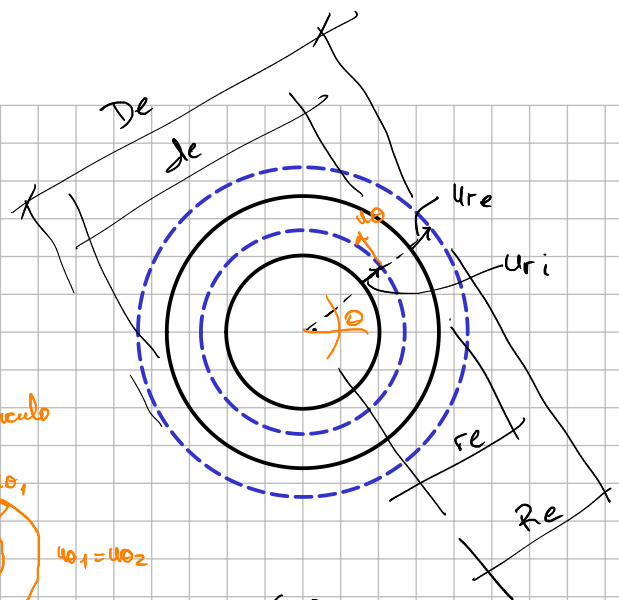


$$\alpha(z) = \tan \alpha(z) = \frac{u_\theta}{r} \Rightarrow u_\theta = \alpha(z) \cdot r$$

$$\frac{\partial u_\theta}{\partial r} = \alpha(z) \cdot \frac{\partial r}{\partial r} = \alpha(z)$$

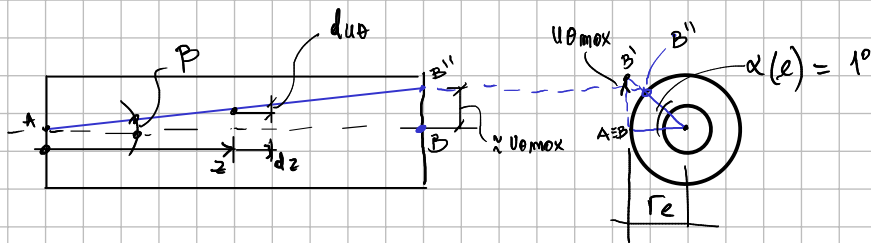
$$\frac{u_\theta}{r} = \frac{\alpha(z) \cdot r}{r} = \alpha(z)$$

$$\epsilon_{r\theta} = \frac{1}{2} \left(\frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r} \right) = \frac{1}{2} \left(\alpha(z) - \alpha(z) \right) = 0$$



• $\epsilon_{z\theta}$

$$\epsilon_{z\theta} = \frac{1}{2} \left(\frac{1}{r} \frac{\partial u_z}{\partial \theta} + \frac{\partial u_\theta}{\partial z} \right) = \frac{1}{2} \frac{\partial u_\theta}{\partial z}$$



$$\tan \beta = \frac{du_\theta}{dz} = \frac{u_{\theta \max}}{r}$$

$$\tan(\alpha(l)) = \frac{u_{\theta \max}}{r_e} \Rightarrow$$

$$\Rightarrow u_{\theta \max} = \tan(\alpha(l)) \cdot r_e$$

$$\frac{du_\theta}{dz} = \frac{\tan(\alpha(l)) \cdot r_e}{r}$$

$$\epsilon_{z\theta} = \frac{1}{2} \frac{du_\theta}{dz} = \frac{1}{2} \frac{\tan(\alpha(l)) \cdot r_e}{r} = \frac{1}{2} \cdot \frac{\tan(1^\circ) \cdot \frac{6 \text{ cm}}{2}}{60 \text{ cm}} = 4,36 \times 10^{-4}$$

• ϵ_{zr}

$$\epsilon_{zr} = \frac{1}{2} \left(\frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r} \right) = 0$$

